

Scientific Machine Learning (SciML)

Module 3: Deep Operator Networks

Topics

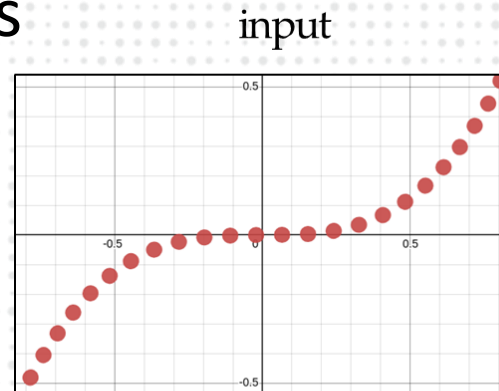
- Review of SNOs
- Deep Operator Networks (DeepONets)
- Implementation of DeepONets

SNOs

- Learn relation between input and output
- Supervised learning (requires data)
- Requires a basis
- Convolution based updates
- Gives resolution and parameter invariance

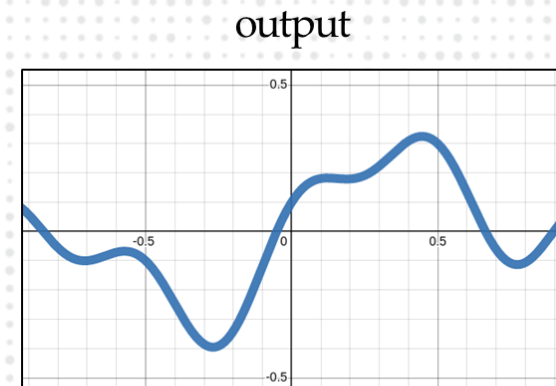
Deep Operator Networks

- Learn $G: a(x) \mapsto u(x)$
- Supervised learning (require data)
- Input:
 - Discrete sample points
 - Coordinates of sample points
- Output:
 - Continuous function



The input function is available at limited points

Operator Network



The output function is learned for all y

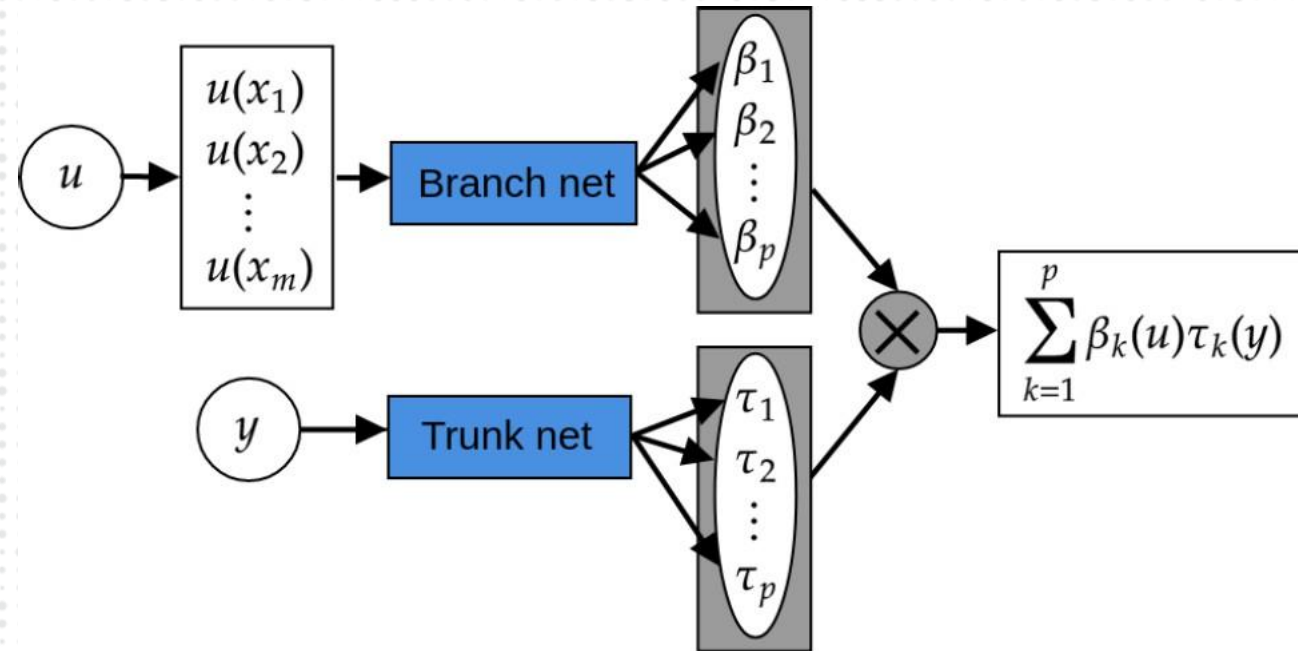
Universal Approximation Theorem

- Any nonlinear operator D can be approximated arbitrarily well by a neural network with at least one hidden layer

$$N(u)(y) = \sum_{k=1} \beta_k(u) \tau_k(y)$$

DeepONets

- Input values are passed to branch
- Input locations (sensors) are passed to trunk
- $u(y)$ is calculated by multiplying branch and trunk outputs



Lu, Lu, Pengzhan Jin, and George Em Karniadakis. "Deeponet: Learning nonlinear operators for identifying differential equations based on the universal approximation theorem of operators." *arXiv preprint arXiv:1910.03193* (2019).

1D Dynamic System

- $s_x = g(s(x), u(x), x)$
- Domain: $\Omega = [0,1]$
- IC: $s(0) = 0$
- Input: $a(x) = [u(x), X]$
- Output: $s(x)$

Live

- Complete discussion on FNOs
 - Review of the code
 - Challenges and limitations of FNOs
- DeepONet implementation
- Challenges and limitations of DeepONet

Comparison

Aspect	DeepONet	FNO	PINN
Learning type	Data-driven (supervised)	Data-driven (spectral)	Physics-informed (unsupervised)
Input–output	Function → Function	Function → Function	PDE residual minimization
Data requirement	High	Moderate	Low / none
Scalability	Poor in high-dim	Excellent (Fourier conv)	Poor (collocation scaling)
Physics usage	None (unless added via loss)	None (unless hybridized)	Explicit (via PDE loss)
Extrapolation	Weak	Moderate–good	Strong (same PDE)
Training speed	Fast (if data available)	Fast	Slow
Interpretability	Moderate	Low	High (physics-based)
Best for	Small/medium-scale operator regression	Large-scale PDE surrogates	Solving PDEs with limited data

THANK YOU